ABEEHA BAIG

AI Engineer | LLM Agents, RAG & Document AI (OCR / Computer Vision)

Karachi, Pakistan

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PROFILE

AI Engineer leading the AI work at Qordata, with 3 years shipping production LLM agents, RAG and document AI for pharma-compliance, healthcare and fintech. Built SQL-agent assistants serving 4 client companies and ~50 daily users, and a hybrid OCR pipeline processing ~400 documents a day. Python, LangGraph, FastAPI, Docker, Google Cloud.

WORK EXPERIENCE

AI Innovation Engineer | *Qordata – Karachi, Pakistan* July 2025 – Present

- Lead the AI work across multiple concurrent projects in a 5–7-person team, shipping every service to production on Google Cloud (Vertex AI, Cloud Run, CI/CD on GCP) as Docker containers with React/TypeScript front ends.
- Built a pharma-compliance agent in which SQL agents fetch answers from the compliance database and an LLM layer generates the response, now serving 4 client companies (5–7 users each) and 10–20 internal users.
- Architected agentic RAG workflows for real-time pharmaceutical compliance monitoring that automate web search and comparison of drug data.
- Shipped a LangGraph chatbot with SQL-agent tools that answers product-change questions in plain English, used by ~50 people daily and replacing manual inspection and analysis of charts and reviews.
- Built a hybrid document-AI pipeline that processes ~400 documents a day (4 batches of ~100), combining AWS Textract, PaddleOCR and RapidOCR with Gemini Document AI and image preprocessing (deskewing, adaptive thresholding, sharpening) to reach ~80% extraction accuracy; added YOLOv8/v9 detectors that flag scratch-outs and whiteouts on paper documents for human review, and Gemini 3 Flash / 3.1 Flash Lite rule classification with structured outputs, reaching near-perfect accuracy on typed-document rules.

Machine Learning Engineer | *PixelEdge Solutions (Pvt.) Limited – Karachi, Pakistan* Oct 2023 – June 2025

- Designed and deployed LangChain agents that resolve multi-step user queries across healthcare and fintech client products.
- Improved chatbot retrieval quality by building modular and advanced RAG pipelines (HyDE, step-back prompting, re-ranking) and benchmarking vector databases with HNSW (approximate) vs flat (exact) indexing.
- Built SQL- and Pandas-agent tools plus regex/NLP extraction pipelines that let LLMs query and reason over financial datasets, turning raw transaction data into natural-language insights and formatted Excel reports.
- Segmented customer reviews with DBSCAN clustering and fed the clusters into a DataFrame for per-group sentiment analysis, turning unstructured feedback into segment-level insights.
- Shipped Flask and FastAPI REST services integrating OpenAI, Claude, Gemini, DeepSeek and LLaMA for healthcare and finance use cases, including a GPT-2 model fine-tuned on healthcare data for domain-specific next-word prediction on a pharmaceutical project.

Machine Learning Intern | *Techno-Hacks Edu Tech – Karachi, Pakistan* Aug 2023 – Sep 2023

- Built classical ML models (classification, regression) on real-world datasets.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript

LLMs & Agents: LangChain, LangGraph, agentic RAG (HyDE, step-back prompting, re-ranking, HNSW/flat vector search), SQL/Pandas agents, tool calling, structured outputs; Gemini, OpenAI, Claude, LLaMA

ML, CV & Document AI: PyTorch, TensorFlow, Keras, scikit-learn, YOLOv8/v9, OCR (AWS Textract, PaddleOCR, RapidOCR, Gemini Document AI), image preprocessing, DBSCAN, sentiment analysis

Backend & MLOps: FastAPI, Flask, REST APIs, Docker, CI/CD, Google Cloud (Vertex AI, Cloud Run, Cloud Storage, Firestore), Git, React

Data: Pandas, NumPy, regex/NLP extraction, web scraping (BeautifulSoup, Selenium)

EDUCATION

Master of Science – Data Science | NEDUET | Jan 2025 – expected Dec 2026 | CGPA: 3.33/4.00

Bachelor of Engineering – Avionics | KIET | Jan 2020 – Dec 2023 | CGPA: 3.56/4.00

LANGUAGES

English: C1 (university English proficiency certificate)

COURSES & CERTIFICATIONS

- OCI Generative AI Professional Certification — Oracle University
- Generative AI with LLMs — Coursera
- AI Agents — Hugging Face
- Building Systems with ChatGPT API and Gemini API — DeepLearning.AI
- n8n for Automation — n8n Academy

PERSONAL PROJECTS

Modular RAG-Based Chatbot for Oracle Database Security Q&A — *github.com/abeeha-baig/RAG-ODS*
RAG Q&A system (LangChain, Streamlit) that retrieves from embedded PDF documentation to generate grounded conversational answers.

Lifestyle Agent Using LangChain and Firecrawl — *github.com/abeeha-baig/Lifestyle-coach-agent*
Wellness/productivity agent that scrapes articles (Medium, Healthline) with Firecrawl and summarises them with GPT-4o-mini.

Multi-Agent Fitness & Macros Coach *(in progress)* — *github.com/abeeha-baig/multi-agents*
Multi-agent LangGraph system for personalised fitness guidance and macro tracking.